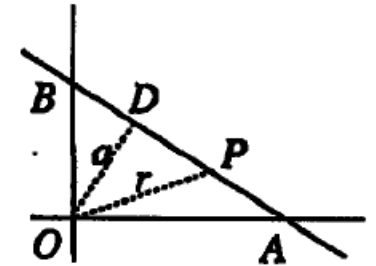


MA1521 L1 Week 03 Mon 20260824

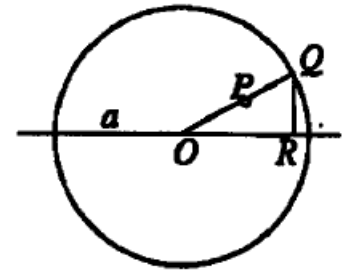
- (i) Find the polar equation for the line intersecting the positive x - and y -axes respectively at A and B , and having perpendicular distance a from the origin.

(Let $\alpha = \angle DOA$; use the right triangle DOP to get the equation connecting r, θ, α and a .



- (ii) Convert your polar equation to the usual rectangular equation involving A and B , by using trigonometry.

In the accompanying figure, the point Q moves around the circle of radius a centered at the origin; QR is a perpendicular to the x -axis. P is a point on ray OQ such that $|QP| = |QR|$: P is the point inside the circle in the first two quadrants, but outside the circle in the last two quadrants.



- (i) Sketch the locus of P ; the locus is called a *cardioid*
- (ii) find the polar equation of this locus.

The point P moves in a locus so that the product of its distances from the two points $Q : (-a, 0)$ and $R : (a, 0)$ is constant. Assuming the locus of P goes through the origin, determine the value of the constant, and derive the polar equation of the locus of P .

(Work with the squares of the distances, rather than the distances themselves, and use the law of cosines; the identities $(A+B)(A-B) = A^2 - B^2$ and $\cos 2\theta = 2\cos^2 \theta - 1$ simplify the algebra and produce a simple answer at the end. The resulting curve is a *lemniscate*,

For each of the following,

- (i) give the corresponding equation in rectangular coordinates;
- (ii) draw the graph; indicate the direction of increasing θ .

a) $r = \sec \theta$ b) $r = 2a \cos \theta$ c) $r = (a + b \cos \theta)$ d) $r = a/(b + c \cos \theta)$

The point P moves with constant speed v in the direction of the constant vector $a\mathbf{i} + b\mathbf{j}$. If at time $t = 0$ it is at (x_0, y_0) , what is its position vector function $\mathbf{r}(t)$?

Describe the motions given by each of the following position vector functions, as t goes from $-\infty$ to ∞ . In each case, give the xy -equation of the curve along which P travels, and tell what part of the curve is actually traced out by P .

a) $\mathbf{r} = 2\cos^2 t\mathbf{i} + \sin^2 t\mathbf{j}$ b) $\mathbf{r} = \cos 2t\mathbf{i} + \cos t\mathbf{j}$ c) $\mathbf{r} = (t^2 + 1)\mathbf{i} + t^3\mathbf{j}$

The cycloid is the curve traced out by a fixed point P on a circle of radius a which rolls along the x -axis in the positive direction, starting when P is at the origin O . Find the vector function OP ; use as variable the angle θ through which the circle has rolled.

(Hint: begin by expressing OP as the sum of three simpler vector functions.)

A bow-and-arrow hunter walks toward the origin along the positive x -axis, with unit speed; at time 0 he is at $x = 10$. His arrow (of unit length) is aimed always toward a rabbit hopping with constant velocity $\sqrt{5}$ in the first quadrant along the line $y = 2x$; at time 0 it is at the origin.

- a) Write down the vector function $\mathbf{A}(t)$ for the arrow at time t .
- b) The hunter shoots (and misses) when closest to the rabbit; when is that?